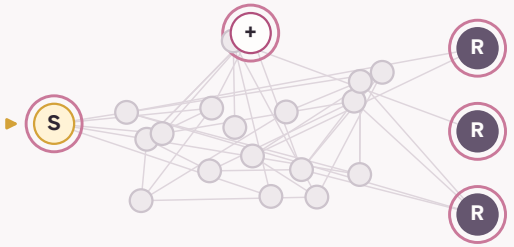
Predictive risk and recording decisions

in partially observed neural networks

Five recordings to predict a held-out stimulus response

1 Record both ends

1 stimulus + 3 readouts + 1 extra site



100-unit network (schematic)

98.13%

lower response prediction error

vs five ordinary recording sites

40 new networks | 95% CI: 93.94-99.19% reduction

2 Explain the benefit

Exact finite-response factorization

Input → Recurrent → Readout

Largest error contraction at input and readout

99.08%

smaller squared response-range width

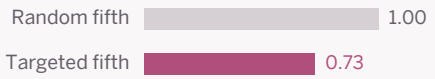
Responses compatible with the calibration fit

40 networks | 95% CI: 96.10-99.62% reduction

3 Acquire internal information

Keep four endpoints; choose the fifth site

Target four recurrent gain combinations



27.15%

smaller squared internal-range width

Along one prespecified internal direction

40 new networks | Adjusted CI: 12.69-43.18%

Calibrate the endpoints, then target the remaining internal information.

Simulated networks with known connectivity and uncertain gains. Internal-range interval adjusted for three comparisons.